



AstraPay Payment Profitability Diagnostic

Full Presentation Transcript — 13 Slides

This is the complete speaking script for every slide in `AstraPay_Complete_Diagnostic.pptx`. The same text is also embedded as speaker notes inside the PowerPoint file (visible in Presenter View or the Notes pane) — this document is a standalone, printable copy of the same script.

Note on Slide 11 (Power BI): the slide itself is intentionally left as an empty placeholder for your own dashboard screenshot — the transcript below still explains the full dashboard design, since you'll speak to these points while showing your actual screenshot.

Slide 1 — Title Slide

Good morning / afternoon. This is the complete walkthrough of the AstraPay payment profitability diagnostic — a five-day analysis covering eight tasks, from problem framing all the way through to a sixty-day action plan. AstraPay is a cross-border payments company operating in India, Singapore, and the UAE. Over the first half of 2025, payment volume grew sharply, but the profit earned on each successful transaction collapsed. That paradox — more business, less money per transaction — is what this entire diagnostic was built to explain. We worked with ten source datasets totaling fifty-six thousand eight hundred and twenty-nine rows, built a trusted analytical model from scratch, tested six competing explanations against real data, and arrived at a five-action plan with named owners and measurable targets. This presentation walks through every stage of that process and the key numbers behind each finding.

Slide 2 — Task 1 — Problem Framing

Task One was Problem Framing. The executive question we were asked to answer was: why is payment contribution profit per successful transaction falling even though payment volume is increasing, and what should AstraPay change in the next sixty days? Before touching any data, we defined the executive KPI precisely: contribution profit per successful transaction, in US dollars, equal to merchant fee revenue plus any FX effect, minus route processing cost, minus fraud and chargeback cost, divided by the count of successful transactions. We also defined six supporting KPIs — payment volume, success rate, gross payment value, processing cost per transaction, fraud and chargeback cost per transaction, and FX impact — because looking only at the headline number would tell us THAT something was wrong but not WHERE. Four stakeholders each proposed a different explanation, and we structured these as a MECE hypothesis set — mutually exclusive, collectively exhaustive — covering four different lenses on the same question: the CFO believed merchant pricing had eroded; the Head of Payments believed routing and processing costs had increased; the Risk Head believed fraud controls were causing false declines; and the Data Head believed the reporting itself could not be trusted. For each hypothesis we defined in advance what evidence would confirm it and what evidence would reject it — this discipline matters because it stops an analyst from unconsciously looking only for evidence that confirms what they already believe.

Slide 3 — Task 2 — Data Contract

Task Two built the Data Contract — a formal agreement covering every one of the ten source tables: what grain each row represents, who owns it, how it joins to other tables, and what validation rules it must satisfy. The single most important concept here is grain — what does one row actually mean. For example, `transaction dot c-s-v` has a grain of one row per payment attempt, but `fraud_decision` has a grain of one row per fraud rule evaluated per transaction, meaning a single transaction can have up to three rows in that table, averaging one point four nine rows per transaction. We identified six specific joins in this dataset that are dangerous if done naively. The worst one: joining `transaction` to `fraud_decision` without pre-aggregating first inflates the total payment amount by forty-nine point eight percent — we proved this by running the join both ways and measuring the actual dollar difference, not just describing the risk theoretically. We also documented exactly

which date field should drive each calculation — `created_at` for volume and activity metrics, because that's when the customer actually attempted to pay; `settlement_date` for realized revenue and its currency conversion, because that's when the money and the exchange rate actually crystallize. This distinction turned out to be critical later, because Route R3's processing fee stepped up on the first of April, and using the wrong date for that lookup would silently cost transactions at the wrong rate.



Slide 4 — Task 3 — Data Quality (1/2): Blocking Defects

Task Three profiled all fifty-six thousand eight hundred twenty-nine rows across the ten files against six quality dimensions. Here are all ten Blocking defects in full — the ones that would silently corrupt a calculation if not explicitly handled. D-zero-one: two hundred eight transactions, two point six percent, have a blank merchant ID, breaking merchant-level segmentation for those rows. D-zero-three-b: thirty-five transactions show status success with an amount of exactly zero — an operational capture glitch that must be excluded from any GPV or profit calculation. D-zero-four: twenty-two fraud risk scores fall outside the valid zero-to-one range, all traced to one model version, fraud v four beta. D-zero-eight: sixty-six merchant-month combinations in the risk snapshot table are duplicated, four point seven six percent of the expected combinations, which would fan out any join to that table. D-zero-nine: three hundred fifty-two transactions have two settlement rows instead of one, five percent of real successful transactions — summing naively double-counts revenue for these. D-ten: `fraud_decision` averages one point four nine rows per transaction, eleven thousand nine hundred fifty-three rows for eight thousand transactions — the exact mechanism behind the forty-nine point eight percent inflation risk. D-eleven: thirty-six transactions are marked successful while the fraud engine also logged a Decline for the same transaction — a genuine contradiction that corrupts fraud-effectiveness metrics if not flagged. D-sixteen: two hundred thirty-two successful transactions, three point three one percent, have no settlement record yet and must never be treated as zero revenue. D-seventeen: seventy-one settlement rows, worth about seventeen thousand and forty-four dollars combined, reference a transaction that doesn't exist anywhere in our data. And D-eighteen is the formal, measured proof of D-ten: running the join naively versus correctly shows an actual plus forty-nine point eight percent inflation in the real INR total, not just a theoretical risk.

Slide 5 — Task 3 — Data Quality (2/2): Warning, Informational & Pass

The remaining nine findings. Four Warnings: D-zero-three-a, one hundred fifty-seven reversed transactions correctly carry a negative amount — legitimate, but needs sign-aware handling so a naive sum doesn't net them against successful volume. D-twelve, three rows where the risk band label doesn't match what the risk score should produce — a rounding-boundary bug, low volume but worth fixing. D-fourteen, one thousand one hundred seventy-nine payment events, nearly five percent, arrive in the warehouse more than a day after they actually happened — use event time, never ingestion time, for business logic. D-fifteen, the processing latency spike on Routes R3 and R5, five hundred fifty-two events running five point seven times slower during the ten to twenty five March window. Three Informational findings, discovered rather than expected: D-zero-five, Route R5 has six months of cost data but zero transactions ever used it. D-zero-six, every single settlement record in six months of data shows status Settled — none of the other three documented statuses ever appear, meaning that logic path is completely untested. D-zero-seven, the risk band value High never appears once across one thousand three hundred eighty six snapshot rows — a question for the Risk Head about whether the model is calibrated correctly. And finally two Pass results, included deliberately to show what was checked and found clean: D-zero-two, the null processing time on non-terminal events is expected by design, not a defect. And D-thirteen, all one hundred fifty-seven reversed transactions were confirmed to have a negative amount with zero exceptions, which is what makes the D-zero-three-a handling rule reliable in the first place. Two entire tables, customer and merchant, came back completely clean — confirming this was a genuine, thorough investigation rather than a scripted result.

Slide 6 — Task 4 — The Trusted Profitability Model

Task Four turned all of that diagnosis into an actual working model. We built it in three layers. Layer one, staging views, collapse every one-to-many table — `fraud_decision`, `settlement`, `chargeback`, and the risk snapshot — down to exactly one row per transaction before anything is allowed to join to the main transaction table; this structurally eliminates the forty-nine point eight percent fan-out risk rather than just documenting it. Layer two joins everything safely into a single fact table, verified to contain exactly eight thousand rows, matching the original transaction count exactly — proof that no join fanned anything out.



Layer three computes the actual dollar economics: revenue in U-S-D using the settlement exchange rate on the settlement date, processing cost using the route's fee schedule on the date the transaction was created, and a contribution profit figure per transaction. One important correction happened during this build: our first attempt assumed AstraPay absorbed the full disputed amount for every chargeback, and that produced an impossible result — profit was deeply negative in every single month, because a handful of large disputes were wiping out razor-thin margins. We corrected this to the industry-standard treatment: the disputed principal is normally clawed back from the merchant's own settlement, and AstraPay's actual cost is a flat ten-dollar dispute-handling fee per case. Once corrected, the model produced an economically sane, still-declining result. Here is the trusted, verified executive KPI, month by month: in January, profit per successful transaction was twenty-nine cents; it peaked in February at thirty-five cents; then it began falling — twenty-seven cents in March, a cliff to just under ten cents in April, a slight recovery to fifteen cents in May, and just over four cents in June. Over the same period, attempted volume grew from one thousand transactions in January to one thousand six hundred twenty-one in June — sixty-two percent growth. Profit per transaction fell eighty-five percent. That is the paradox, now precisely measured on a model we can trust.

Slide 7 — Task 5 — Root Cause: Route R3

Task Five's root cause investigation starts with routing, because it is the single strongest lens on this decline. Two things happened to Route R3, operated by PayGlobal, at the same time. First, on the first of April 2025, its variable processing fee stepped up from one point two percent to one point six percent of every transaction — a thirty-three percent cost increase overnight, confirmed directly from the route cost data: average fee before April was one point one nine seven percent, after April one point six zero three percent. Second, completely independent of that fee change, R3's share of successful payment volume climbed every single month — from twenty-four point nine percent in January to thirty-seven point nine percent in June — across all three countries. To separate these two effects, we ran a mix-versus-rate decomposition comparing the January-February baseline period against the May-June latest period. The mix effect for R3 — what changed purely because volume moved to it, holding its old economics constant — was actually slightly positive, plus three point two cents. That's an important, non-obvious finding: sending more volume to R3 was not inherently a bad decision under the OLD fee structure. The rate effect — what changed purely because R3's own economics got worse — was minus seventeen point one cents, more than five times larger. In other words, it wasn't the extra volume that hurt profit. It was the fee increase itself, striking a route that was carrying a growing share of the business. Combined, Route R3 alone explains sixty-one point seven percent of the entire measured profit decline. Route R4, the UAE local bank route, adds a further twenty-two point five percent, driven by a similar though smaller pattern — its blended economics fell as UAE volume increasingly shifted toward the pricier R3. Together, these two routes explain more than eighty-four percent of the total decline.

Slide 8 — Task 5 — Segment Deep Dive

Beyond routing, we sliced the same decomposition across merchant category, merchant country, and customer segment to make sure we weren't missing an independent driver. By merchant category, Electronics stands out sharply: its chargeback rate rose from one point three seven percent in January to six point four six percent in June — nearly a five times increase — and Electronics alone explains forty-two point nine percent of the merchant-attributable share of the decline, almost entirely through its own worsening economics rather than any shift in volume mix. Digital Goods shows a similar, slightly smaller pattern, rising from about two percent to over seven percent chargeback rate. By merchant country, UAE actually had the BEST starting economics of the three countries — seventy-two cents average profit per transaction at baseline — yet it explains fifty-five percent of the country-level decline, because an increasing share of UAE's own volume is being routed through the pricier cross-border Route R3, dragging its blended average down. This is the same routing story showing up again through a different lens, which strengthens rather than weakens the routing conclusion. By customer segment, Retail customers explain fifty-two percent of the decline and S-M-E customers a further forty-three percent — together accounting for virtually all of it — while Enterprise-segment customers are comparatively insulated, likely because Enterprise transactions skew toward lower-fee pricing plans and routes. One encouraging counter-example worth noting: the Marketplace merchant category actually improved slightly over the period, a reminder that this is not a uniform, company-wide collapse — it is concentrated in specific, identifiable segments.



Slide 9 — Task 5/6 — Testing the Risk Head's Hypothesis

One of the most important findings in this entire diagnostic is a disproof, not a confirmation. The Risk Head's hypothesis was that tighter fraud controls were causing the profit decline through false declines. We tested this directly. Starting in April 2025, a new fraud scoring model — version four point zero beta — was applied increasingly to High-risk-tier merchants specifically. And the evidence for tightening is completely real: the decline rate for High-risk merchants doubled, from eight point two five percent in January to sixteen point three five percent in May, while every other risk tier stayed essentially flat, between seven and nine percent throughout. So the Risk Head is correct that fraud controls tightened and blocked more payments. But here is the critical test: fraud controls can only affect payments while they're being decided — approve or decline. Once a payment succeeds, the fraud engine has no further influence on its economics. So we asked: do the High-risk transactions that DID succeed show worse profit than other tiers? The answer is no. High-risk tier profit per successful transaction went from thirty-three point two cents in January to twenty-two point seven cents in June — a real decline, but a modest one. The Low-risk tier — which this new fraud model never touched at all — fell from thirty point one cents all the way to negative four point four cents, a steeper collapse than the tier that was supposedly being hurt by fraud tightening. If fraud controls were driving the profit decline, we would expect the opposite pattern. We do not see it. The conclusion: fraud tightening is real and it is a genuine volume and success-rate problem worth fixing — but it is not the explanation for the specific KPI management asked about. The Risk Head is right about a real problem, and wrong about which number it explains. We ran the same kind of direct test on the March processing latency incident — routes R3 and R5 ran five point seven times slower than normal for sixteen days in March — and found average profit DURING that slow window was actually twenty-one point seven cents, higher than the four point one cents seen on Route R3 in the months outside the incident. The timing doesn't support latency as a profit driver either — the incident happened in March, before the April fee change, so it's simply the wrong period to explain a decline that deepens afterward.

Slide 10 — Task 6 — Full Diagnosis: All Six Drivers Tested

Task Six brought every candidate explanation together into one formal diagnosis, testing all six possible drivers named across the case brief and the four stakeholders. Processing latency: real, a five point seven times slowdown, but wrong timeline — proven not to explain the decline. Fraud controls: real, decline rate doubled for High-risk merchants, but disproven as an explanation for the profit-per-transaction KPI specifically — it's a volume problem. Routing cost and mix: confirmed as the PRIMARY driver, sixty-one point seven percent of the total decline, dominated by Route R3's fee increase. Chargeback drift in Electronics and Digital Goods: confirmed as a SECONDARY, independent driver, forty-two point nine percent of the merchant-attributable decline concentrated in those two categories. Merchant pricing: the CFO's concern is real but narrow — Singapore Enterprise-plan merchants had their fee cut from two point six percent to one point five percent starting February 2025, a genuine, deliberate business decision — but it touches only one plan in one country, far too small to explain a three-country, six-month, eighty-five percent collapse, and no broader undocumented pricing erosion was found anywhere else. And finally, F-X rate selection: this is fundamentally different from the other five — it is not a business event at all, it is a measurement risk. The dataset carries three different exchange rate types for the same currency and date, and using the wrong one — interbank instead of settlement — can shift computed U-S-D profit by roughly one point one percentage points, large enough to distort a trend if applied inconsistently. This was fixed permanently at the model layer in Task Four by explicitly specifying the settlement rate everywhere revenue is calculated. Two confirmed drivers. Two real problems pointing at the wrong metric. One minor, narrow effect. One measurement risk already corrected. That is the complete diagnosis.

Slide 11 — Task 7 — Executive Dashboard Design (screenshot placeholder)

Task Seven designed the executive Power B-I dashboard, built around a strict rule: every visual reads from a single governed table — the trusted profitability model we built in Task Four — rather than ten raw CSV files joined ad hoc inside the report. That single design choice eliminates the join fan-out risk at the reporting layer entirely, the same way the SQL staging layer eliminated it upstream. The dashboard has four K-P-I cards at the top: contribution profit per successful transaction, attempted volume with growth versus January, success rate, and a settlement-pending percentage card that keeps the data-trust caveat visible on the same screen as the headline number, not buried in a footnote. Below that, two pages. The Diagnostic page has two visuals: a combined line-and-bar chart showing volume climbing while profit per transaction falls — the paradox in one picture — and a waterfall chart bridging from the two-month baseline profit of thirty-two point one three cents down to the two-month latest figure of nine point six eight cents, broken out route by route, so the sixty-one point seven percent R3

contribution is visually unmistakable. The Action page also has two visuals: a route-by-month drill matrix that a decision-maker can click into — selecting Route R3 and June, for example, instantly cross-filters every K-P-I card above to just that slice — and a sixty-day action tracker showing current value versus target for each of the five recommended actions. Throughout, every non-additive measure — success rate, profit per transaction, any percentage — is calculated as a ratio of two sums, never as an average of a pre-computed percentage column, because averaging percentages across months of very different volume would silently misstate the true trend. [Presenter note: this slide intentionally shows a placeholder — replace the dashed box with your own Power BI screenshot before presenting.]

Slide 12 — Task 8 — 60-Day Action Plan

Task Eight translates every finding into five concrete actions, each with a named owner, a day range, an expected impact, a risk, and a measurable success metric. Action one, days one through thirty, owned jointly by the Head of Payments and the C-F-O: renegotiate or cap Route R3's variable fee back toward one point two percent — this is the single biggest lever, potentially recovering eight to twelve cents per transaction. Action two, days one through twenty, Head of Payments: rebalance ten to fifteen percent of volume away from R3 toward the other routes immediately, even before renegotiation completes, targeting R3's share back below thirty percent. Action three, days one through forty-five, Data Head and Payments Engineering jointly: fix all ten Blocking data quality defects identified in Task Three, prioritizing the settlement and fraud fan-out issues first — this is explicitly called out as a prerequisite, because none of the other four actions can be reliably measured on data that is still producing forty-nine percent inflated totals. Action four, days twenty through sixty, Risk Head and Merchant Success: launch a dispute-prevention program specifically for Electronics and Digital Goods merchants, targeting a return to the January baseline chargeback rate — with the caveat that chargebacks have roughly a sixty-day filing lag, so results won't be visible until day ninety. Action five, days thirty through sixty, Risk Head: formally review the fraud v four beta model through a proper A-B test — not a blanket rollback, since rolling back could increase actual fraud losses — targeting the High-risk decline rate back under ten percent while confirming no increase in real fraud.

Slide 13 — Summary — Key Takeaways

To close, here is the whole diagnostic in one page. The starting paradox: payment volume grew sixty-two percent while contribution profit per successful transaction fell eighty-five percent, from twenty-nine cents in January to just over four cents in June. Getting to that trustworthy number required fixing nineteen data quality defects first, ten of them severe enough to silently corrupt any calculation built on top of them — including a forty-nine point eight percent inflation risk from a single careless join. Once the data could be trusted, root-cause analysis showed the decline is concentrated, not universal: Route R3 alone explains sixty-one point seven percent of it, driven overwhelmingly by an April fee increase rather than the volume growth itself; a further twenty-two point five percent comes from Route R4; and a rising chargeback rate in Electronics and Digital Goods merchants explains forty-two point nine percent of the merchant-level share. Two other stakeholder theories — fraud controls and processing latency — were both confirmed as real operational issues, but both were proven, with direct evidence, to explain a different problem than the one being asked about. Pricing was real but too narrow to matter at this scale, and the F-X measurement risk was corrected permanently in the model. The result is a five-action, sixty-day plan with named owners and hard success metrics, starting with the routing fee and mix, running the data quality fixes in parallel as a prerequisite for everything else, and following with the chargeback and fraud reviews on realistic timelines. The core lesson of this entire exercise: all four stakeholders were describing a real piece of the truth. The job of this diagnostic was not to declare a winner, but to quantify exactly how much of the decline each explanation actually accounts for, and to build a plan around the answers, not the assumptions.